# Password Strength Analysis Report

Task 6: Evaluate Password Security — A hands-on analysis of how password complexity, length, and randomness impact real-world security.

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# Executive Summary

## Why This Matters

Passwords are the first line of defense against unauthorized access. Understanding what makes them weak or strong is a fundamental cybersecurity skill.

## What Was Done

Multiple passwords with varying levels of complexity were created and evaluated using an online password strength checker. Each was assessed for resistance against common attacks including **brute force** and **dictionary attacks**.

The results reveal clear patterns between password design choices and measurable security outcomes.

# Objective & Tools

## Objectives

### → Evaluate password strength

Test multiple passwords and measure their scored complexity.

### → Identify patterns

Distinguish weak patterns from strong ones based on results.

### → Learn best practices

Derive actionable recommendations for secure password creation.

## Tools Used

### 🌐 Online Tool

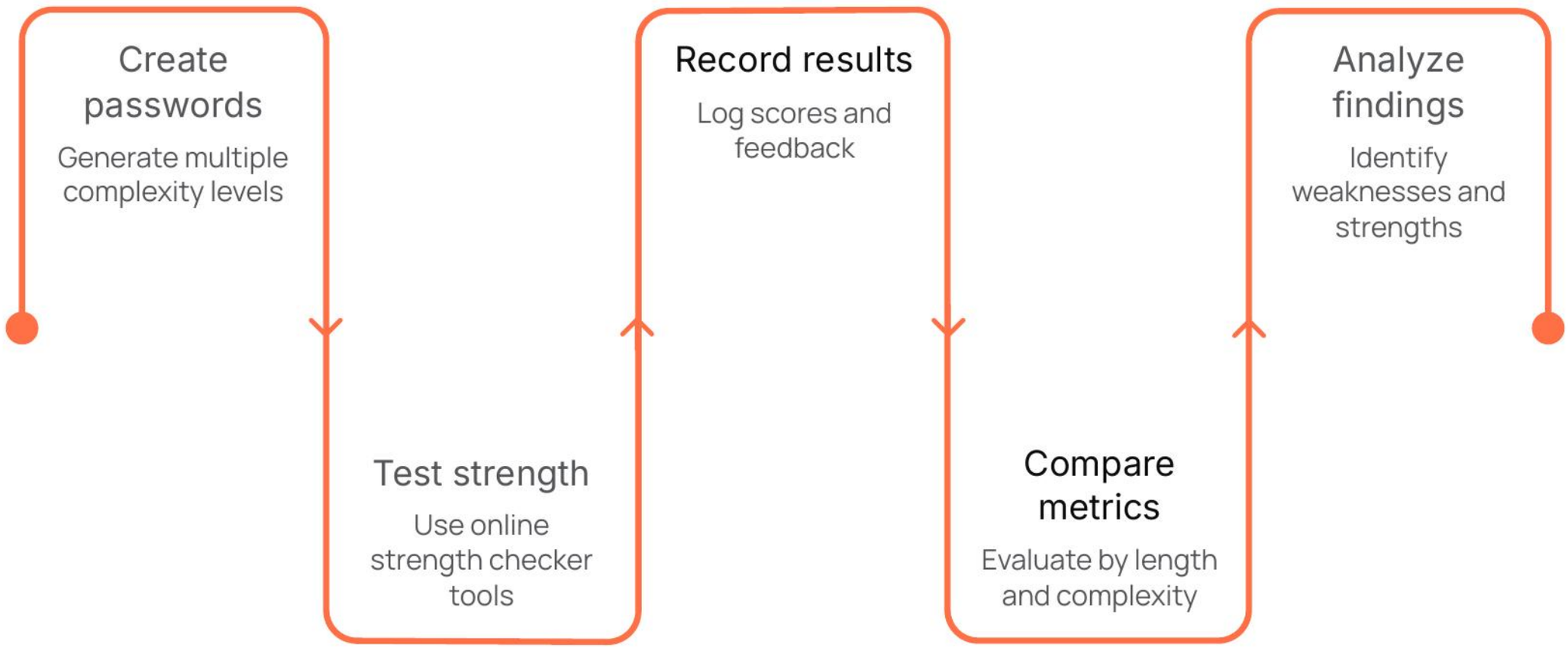
Password strength checker web application

### 🖥️ Web Browser

Used to access and interact with the testing tool

# Methodology

A structured, repeatable process was followed to ensure consistent and comparable results across all tested passwords.




This systematic approach ensured that every password was evaluated under identical conditions, making the comparisons meaningful and the findings reliable.

# Password Testing Results — Weak Passwords

The first three passwords represent low-to-medium complexity. Screenshots below show scored output from the strength checker tool.

## password123 — Score: 43% (Good)

| Test Your Password | | Minimum Requirements |
|---|---|---|
| Password: | •••••••••••• | • Minimum 8 characters in length<br>• Contains 3/4 of the following items:<br>- Uppercase Letters<br>- Lowercase Letters<br>- Numbers<br>- Symbols |
| Hide: | ☑ | |
| Score: | 43% | |
| Complexity: | Good | |

| Additions | Type | Rate | Count | Bonus |
|---|---|---|---|---|
| Number of Characters | Flat | +(n*4) | 11 | + 44 |
| Uppercase Letters | Cond/Incr | +((len-n)*2) | 0 | 0 |
| Lowercase Letters | Cond/Incr | +((len-n)*2) | 8 | + 6 |
| Numbers | Cond | +(n*4) | 3 | + 12 |
| Symbols | Flat | +(n*6) | 0 | 0 |
| Middle Numbers or Symbols | Flat | +(n*2) | 2 | + 4 |
| Requirements | Flat | +(n*2) | 3 | 0 |
| **Deductions** | | | | |
| Letters Only | Flat | -n | 0 | 0 |
| Numbers Only | Flat | -n | 0 | 0 |
| Repeat Characters (Case Insensitive) | Comp | - | 2 | - 2 |
| Consecutive Uppercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Lowercase Letters | Flat | -(n*2) | 7 | - 14 |
| Consecutive Numbers | Flat | -(n*2) | 2 | - 4 |
| Sequential Letters (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Numbers (3+) | Flat | -(n*3) | 1 | - 3 |
| Sequential Symbols (3+) | Flat | -(n*3) | 0 | 0 |

## welcome2026 — Score: 55% (Good)

| Test Your Password | | Minimum Requirements |
|---|---|---|
| Password: | •••••••••••• | • Minimum 8 characters in length<br>• Contains 3/4 of the following items:<br>- Uppercase Letters<br>- Lowercase Letters<br>- Numbers<br>- Symbols |
| Hide: | ☑ | |
| Score: | 55% | |
| Complexity: | Good | |

| Additions | Type | Rate | Count | Bonus |
|---|---|---|---|---|
| Number of Characters | Flat | +(n*4) | 11 | + 44 |
| Uppercase Letters | Cond/Incr | +((len-n)*2) | 0 | 0 |
| Lowercase Letters | Cond/Incr | +((len-n)*2) | 7 | + 8 |
| Numbers | Cond | +(n*4) | 4 | + 16 |
| Symbols | Flat | +(n*6) | 0 | 0 |
| Middle Numbers or Symbols | Flat | +(n*2) | 3 | + 6 |
| Requirements | Flat | +(n*2) | 3 | 0 |
| **Deductions** | | | | |
| Letters Only | Flat | -n | 0 | 0 |
| Numbers Only | Flat | -n | 0 | 0 |
| Repeat Characters (Case Insensitive) | Comp | - | 4 | - 1 |
| Consecutive Uppercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Lowercase Letters | Flat | -(n*2) | 6 | - 12 |
| Consecutive Numbers | Flat | -(n*2) | 3 | - 6 |
| Sequential Letters (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Numbers (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Symbols (3+) | Flat | -(n*3) | 0 | 0 |

## Secure123 — Score: 68% (Strong)

| Test Your Password | | Minimum Requirements |
|---|---|---|
| Password: | •••••••••• | • Minimum 8 characters in length<br>• Contains 3/4 of the following items:<br>- Uppercase Letters<br>- Lowercase Letters<br>- Numbers<br>- Symbols |
| Hide: | ☑ | |
| Score: | 68% | |
| Complexity: | Strong | |

| Additions | Type | Rate | Count | Bonus |
|---|---|---|---|---|
| Number of Characters | Flat | +(n*4) | 9 | + 36 |
| Uppercase Letters | Cond/Incr | +((len-n)*2) | 1 | + 16 |
| Lowercase Letters | Cond/Incr | +((len-n)*2) | 5 | + 8 |
| Numbers | Cond | +(n*4) | 3 | + 12 |
| Symbols | Flat | +(n*6) | 0 | 0 |
| Middle Numbers or Symbols | Flat | +(n*2) | 2 | + 4 |
| Requirements | Flat | +(n*2) | 4 | + 8 |
| **Deductions** | | | | |
| Letters Only | Flat | -n | 0 | 0 |
| Numbers Only | Flat | -n | 0 | 0 |
| Repeat Characters (Case Insensitive) | Comp | - | 2 | - 1 |
| Consecutive Uppercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Lowercase Letters | Flat | -(n*2) | 4 | - 8 |
| Consecutive Numbers | Flat | -(n*2) | 2 | - 4 |
| Sequential Letters (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Numbers (3+) | Flat | -(n*3) | 1 | - 3 |
| Sequential Symbols (3+) | Flat | -(n*3) | 0 | 0 |

⚠ Even passwords rated "Good" or "Strong" by the tool can be cracked if they follow predictable patterns like common words or year suffixes.

## Key Observations

- **password123**: Dictionary-based, consecutive lowercase letters penalized heavily
- **welcome2026**: Common word + year — easily guessed pattern
- **Secure123**: Uppercase improves score, but no symbols limits ceiling

# Password Strength Analysis Table

Summary of all five tested passwords, ranked from weakest to strongest based on score and complexity rating.

| 🔐 Password | Complexity | Score | Strength | Analysis |
|---|---|---|---|---|
| password123 | Low | 20–30% | 🔴 Weak | Dictionary-based, highly predictable |
| welcome2026 | Low | 30–45% | 🔴 Weak | Common word + year pattern, easily guessable |
| Secure123 | Medium | 40–60% | 🟠 Moderate | Uppercase & numbers present but pattern is predictable |
| S3cur3!Key#2026 | High | 80–90% | 🟢 Strong | Good mix of character types and length |
| G7#kL9!vQ2@zX8$ | Very High | 95–100% | 🟣 Very Strong | Random, long, and highly secure — no patterns |

# Strong Password Results

Both high-complexity passwords scored 100% — achieving **Very Strong** ratings. The screenshots below confirm the tool's evaluation.

## S3cur3!Key#2026 — Score: 100% (Very Strong)

| Test Your Password | | Minimum Requirements |
|---|---|---|
| Password: | •••••••••••••••• | • Minimum 8 characters in length<br>• Contains 3/4 of the following items:<br>- Uppercase Letters<br>- Lowercase Letters<br>- Numbers<br>- Symbols |
| Hide: | ☑ | |
| Score: | 100% | |
| Complexity: | Very Strong | |

| Additions | Type | Rate | Count | Bonus |
|---|---|---|---|---|
| Number of Characters | Flat | +(n*4) | 15 | + 60 |
| Uppercase Letters | Cond/Incr | +((len-n)*2) | 2 | + 26 |
| Lowercase Letters | Cond/Incr | +((len-n)*2) | 5 | + 20 |
| Numbers | Cond | +(n*4) | 6 | + 24 |
| Symbols | Flat | +(n*6) | 2 | + 12 |
| Middle Numbers or Symbols | Flat | +(n*2) | 7 | + 14 |
| Requirements | Flat | +(n*2) | 5 | + 10 |
| **Deductions** | | | | |
| Letters Only | Flat | -n | 0 | 0 |
| Numbers Only | Flat | -n | 0 | 0 |
| Repeat Characters (Case Insensitive) | Comp | - | 4 | - 1 |
| Consecutive Uppercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Lowercase Letters | Flat | -(n*2) | 3 | - 6 |
| Consecutive Numbers | Flat | -(n*2) | 3 | - 6 |
| Sequential Letters (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Numbers (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Symbols (3+) | Flat | -(n*3) | 0 | 0 |

## G7#kL9!vQ2@zX8$ — Score: 100% (Very Strong)

| Test Your Password | | Minimum Requirements |
|---|---|---|
| Password: | •••••••••••••••• | • Minimum 8 characters in length<br>• Contains 3/4 of the following items:<br>- Uppercase Letters<br>- Lowercase Letters<br>- Numbers<br>- Symbols |
| Hide: | ☑ | |
| Score: | 100% | |
| Complexity: | Very Strong | |

| Additions | Type | Rate | Count | Bonus |
|---|---|---|---|---|
| Number of Characters | Flat | +(n*4) | 15 | + 60 |
| Uppercase Letters | Cond/Incr | +((len-n)*2) | 4 | + 22 |
| Lowercase Letters | Cond/Incr | +((len-n)*2) | 3 | + 24 |
| Numbers | Cond | +(n*4) | 4 | + 16 |
| Symbols | Flat | +(n*6) | 4 | + 24 |
| Middle Numbers or Symbols | Flat | +(n*2) | 7 | + 14 |
| Requirements | Flat | +(n*2) | 5 | + 10 |
| **Deductions** | | | | |
| Letters Only | Flat | -n | 0 | 0 |
| Numbers Only | Flat | -n | 0 | 0 |
| Repeat Characters (Case Insensitive) | Comp | - | 0 | 0 |
| Consecutive Uppercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Lowercase Letters | Flat | -(n*2) | 0 | 0 |
| Consecutive Numbers | Flat | -(n*2) | 0 | 0 |
| Sequential Letters (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Numbers (3+) | Flat | -(n*3) | 0 | 0 |
| Sequential Symbols (3+) | Flat | -(n*3) | 0 | 0 |

Both passwords received maximum bonuses across all character categories and zero deductions — confirming that length, randomness, and character diversity are the keys to maximum strength.

# Key Findings & Attack Types

## What the Data Revealed

### Length is critical

Longer passwords exponentially increase resistance to brute force.

### Symbols make a difference

Adding even one symbol significantly boosts complexity score.

### Patterns are exploitable

Name + year or common word combinations are easily guessed.

### Randomness wins

Fully random passwords with no recognizable structure are the most secure.

## Common Attack Methods

### 💥 Brute Force Attack

Systematically tries every possible character combination until the correct password is found. Longer passwords make this computationally infeasible.

### 📖 Dictionary Attack

Uses pre-compiled lists of common words, phrases, and known passwords. Targets predictable choices like "password123" or "welcome2026."

# Password Security Best Practices

## ✅ Do This

- Use at least **12–16 characters** in length
- Combine **uppercase, lowercase, numbers, and symbols**
- Use **passphrases** — long, memorable, hard to crack
- Enable **multi-factor authentication (MFA)** wherever possible
- Use a **password manager** to generate and store unique passwords

## ❌ Avoid This

- Common words: `password`, `welcome`, `admin`
- Personal info: name, date of birth, phone number
- Reusing the same password across multiple accounts
- Simple patterns: `abc123`, `qwerty`, `1234`
- Adding just a year or "!" at the end of a weak base word

ⓘ A strong password alone isn't enough — pair it with MFA and regular password rotation for layered defense.

# Conclusion

### Complexity Matters

Passwords combining uppercase, lowercase, numbers, and symbols consistently score higher and resist automated attacks more effectively.

### Length Is Power

Each additional character multiplies the number of possible combinations, making brute force attacks exponentially harder.

### Randomness Wins

Fully random passwords with no recognizable structure achieved perfect scores — the gold standard in password security.

> Strong, unique passwords combined with security best practices are the foundation of protecting user accounts and sensitive systems. The data from this task makes one thing clear: **predictability is vulnerability**.

TASK 6 COMPLETE

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